

WHAT IS OPERATING SYSTEM

1. Os is the main part of the entire system.
2. It is a intermediate distance between the user and the hardware.
3. and it helps in the task such as
    1. Process management
    2. Memory management
    3. File handling
    4. Device control

TYPES OF OPERATING SYSTEM

It has two types: -

1. Cli (command line interface)
   Example: - Linux
2. GUI (graphical user interface)
   Example: - windows

HISTORY OF UNIX

- In 1964 ------> started project by Bella labs (new jersey)
- In 1969 ------> stopped the project
- In 1969 ------> Restarted by Dennis Ritche and ken Thompson developed a multiuser multitasking os called UNICS

UNICS: Uniplexed Information and Computing Service

- In 1975 ------> its version 6 more popular and multiple companies create their own os versions based on unix

NOTE:

1. UNICS later renamed to UNIX
2. Dennis Ritchie developed the C programming language at Bell Labs in 1972.

UNIX FLAVORS

1. IBM ----------> AIX
2. HP ----------> UX
3. Apple ----------> MacOS
4. Sun Microsystem ----> solaria's

HISTORY OF LINUX

- In 1983 ------> American engineer Richard Stallman started project GNU project
- In 9185 ------> FSF started (free software foundation)
- In 1991 ------> started Programmes like debugger shell, editor
- In 1991 ------> Finnish software engineer Linux Torvalds develops the Kernal

Kernal ----> a core part of operating system

- Kernal + GNU project ----> complete Linux os
- Linux name also came from the engineer Linus Torvalds

LINUX DISTRIBUTIONS

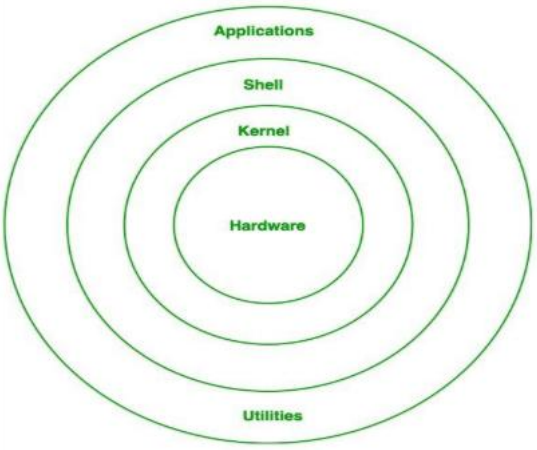
1. RedHat
2. Debian
3. Fedora
4. Others
    1. Centos
    2. Ubuntu
    3. Kali linux
    4. Amazon linux

LINUX CHARACTERSTICS

1. Free and Open source
2. Secure
3. Light weight hardware
4. Portable
5. Multiuser multitasking

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LINUX ARCHITECTURE



Hardware

1. Hardware is a physical component or logical component of any electronic device that Stores the data and run the written code provided by the software.
2. Example: keyboard, mouse, monitor, CPU

Kernal

1. Kernal is a core part of os
2. And has a complete control over everything in the system
3. And its a main layer between the hardware and the Software and performs the task such as process management, memory management and file management

Shell

1. shell is a Linux command line interpreter
2. It provides an interface distance between the user and the kernel
3. it can simply convert the users high level command into the low-level code (to understand the kernel)
4. Example: bash, ksh, zsh, sh

Software

Software is a set of instructions or collection of command that tells the computer how to perform a particular task

Firmware

Firmware is also a set of instructions written in a machine language which are used to operate the hardware

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What is Server

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A server is a system (physical machine) that stores the data, runs applications, and provides services to client machines over a network, usually identified by a static IP address.

Desktop OS vs Server OS

Desktop OS

1. A Desktop Operating System is designed for individual users to perform daily tasks.
2. Example: - windows, ubuntu, macOS

Server OS

1. A Server Operating System is designed to manage resources and serve multiple clients over a network.
2. Example: - Ubuntu server, windows server, Redhat

DEFFERENCE BETWEEN THE WINDOWS / LINUX / UNIX

WINDOWS: -

1.Its a proprietary based (means licensing is required for using windows)
2.it is less secure (more targeted by viruses)
3.it has a heavy hardware (means requires more system resources (RAM/CPU)
4.it is 80% on desktop
5.it is easy to use because of its GUI based

UNIX: -

1.It's a proprietary based (means licensing is required to use)
2.it is more secure
3.it has a lightweight hardware
4.it is 90% on approx. servers
5.it is hard to use because of its CLI based

LINUX: -

1.It's a community based, Free and open-source operating system
2.it is more secure
3.it has a lightweight hardware
4.it is 90% on approx. servers
5.it is hard to use because of its CLI based

WHAT IS LINUX

Linux is a free and open-source kernel that provides the stability, Security, multiuser multitasking capabilities and supports the wide range of hardware's.

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PLATFORMS TO USE LINUX

1. Labex.io
2. Ubuntu (WSL on windows)
3. AWS ec2
4. Virtual Machine (VirtualBox)
